한국폴리텍대학

네트워크 실무

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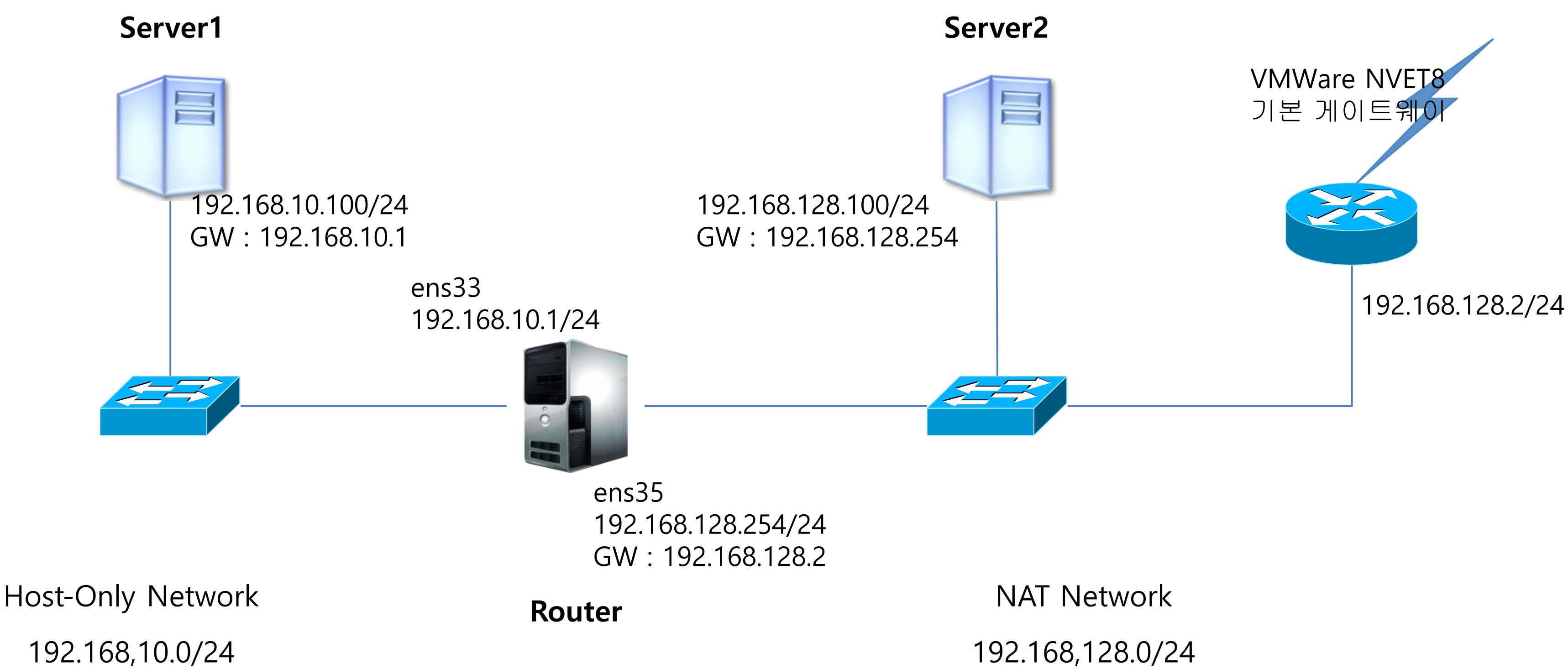
1. LINUX 라우터

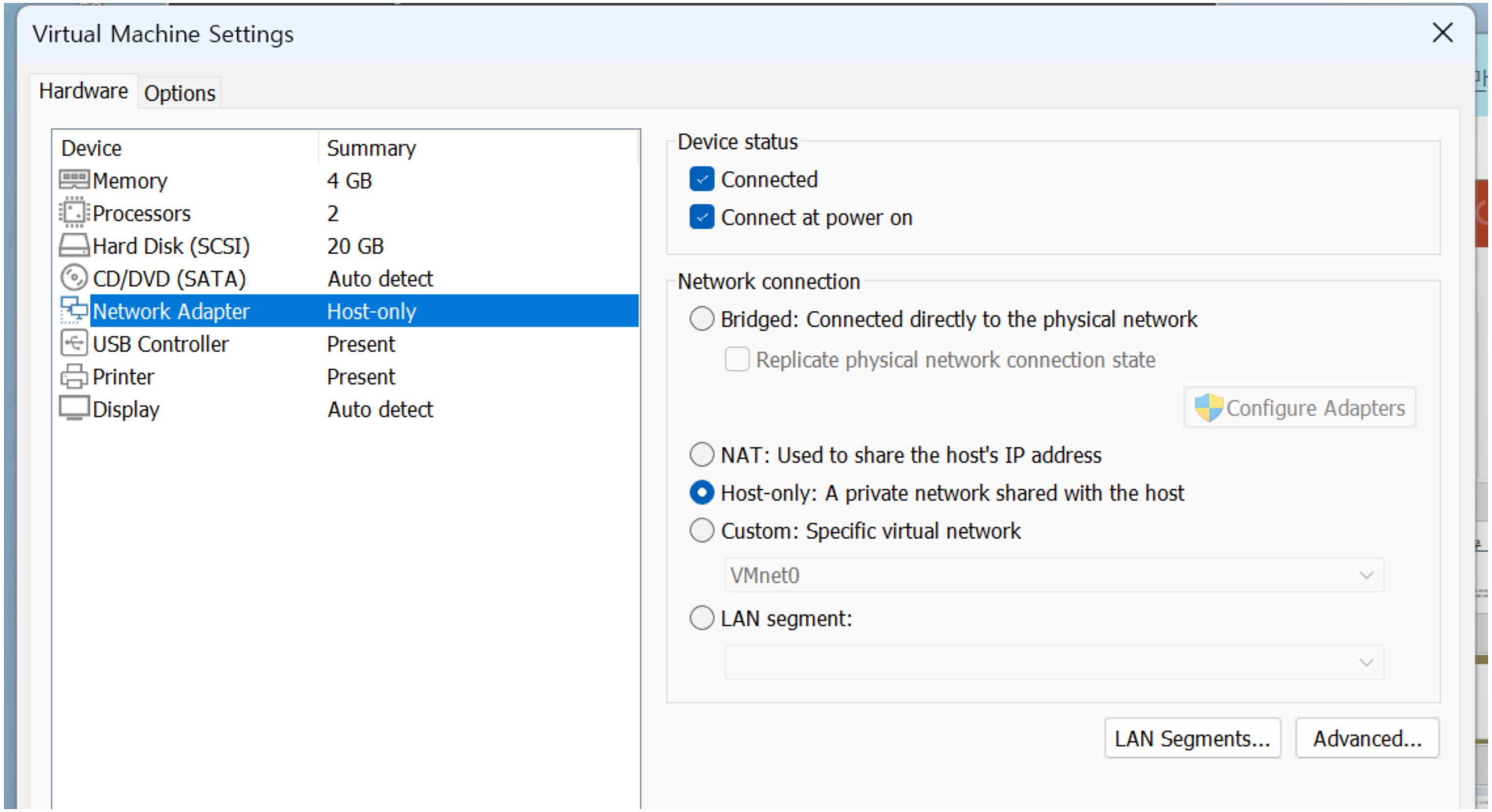
구성도

- 우분투 리눅스로 라우터를 만들어 서로 다른 네트워크간 통신을 할 수 있도록 설정하도록 합니다.



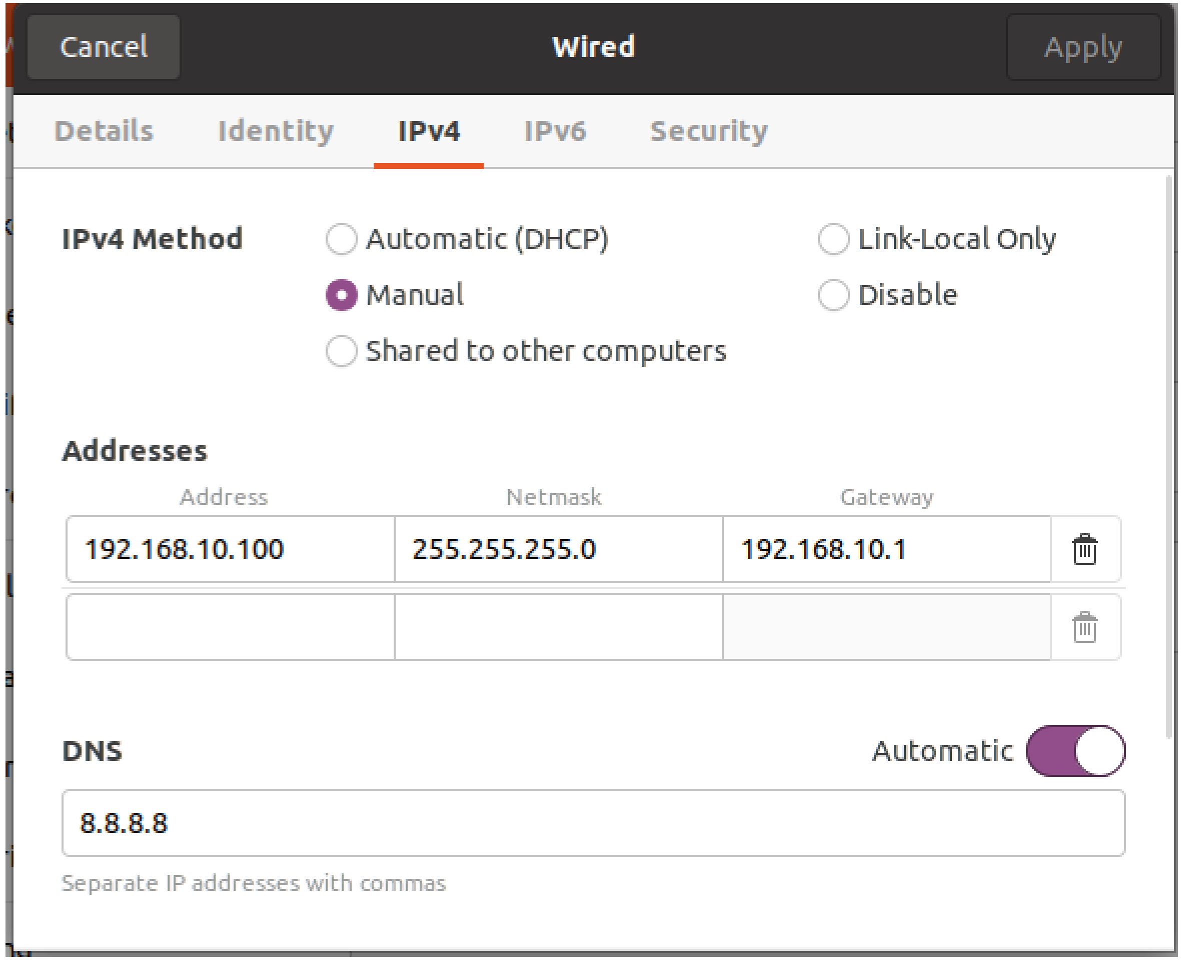
Server1 설정

- Network Adapter : Host-Only



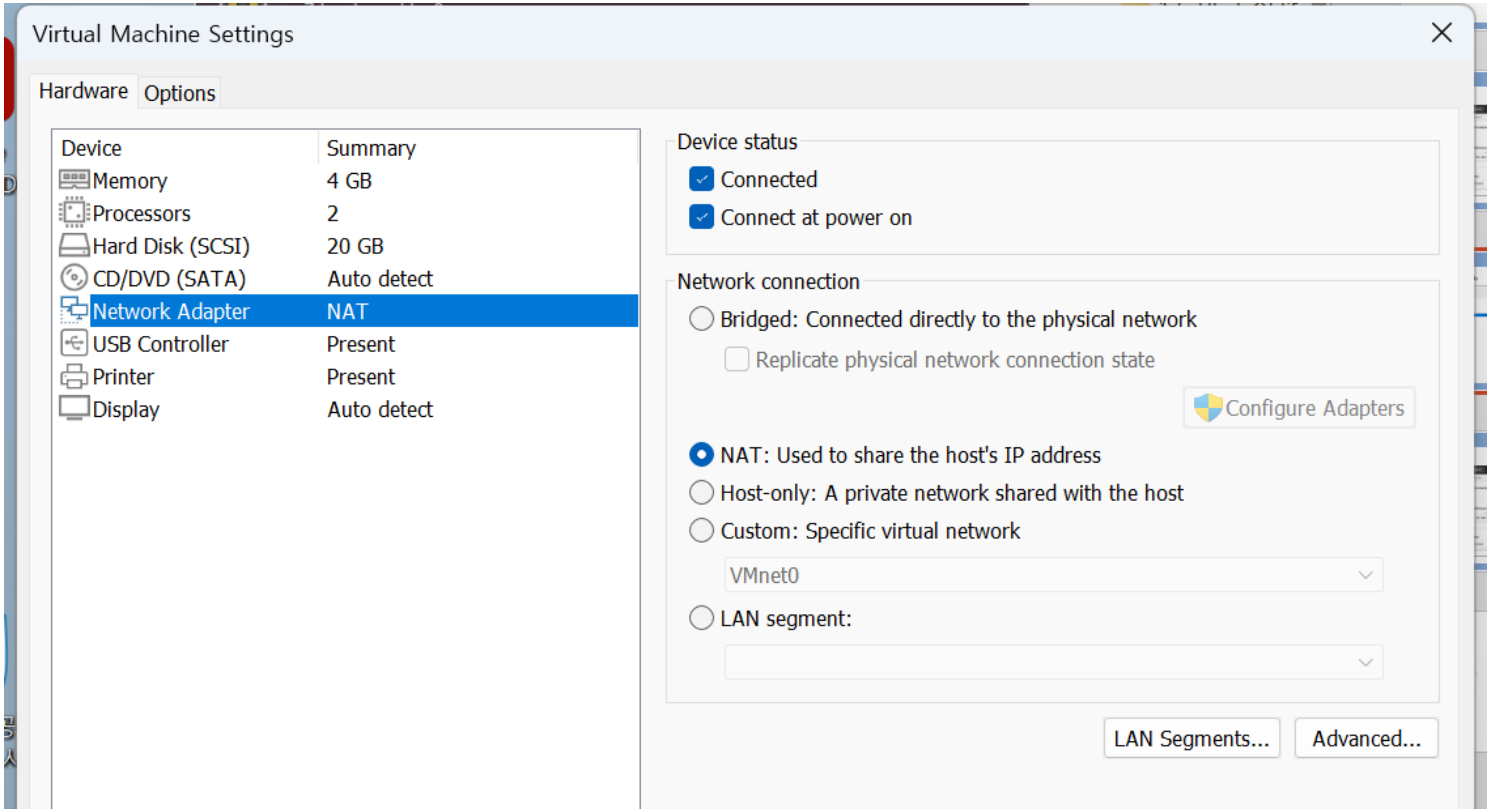
Server1 설정

- Ubuntu Network 설정
  - IPv4 Method : Manual
  - IP : 192.168.10.100
  - Netmask : 24
  - Gateway : 192.168.10.1



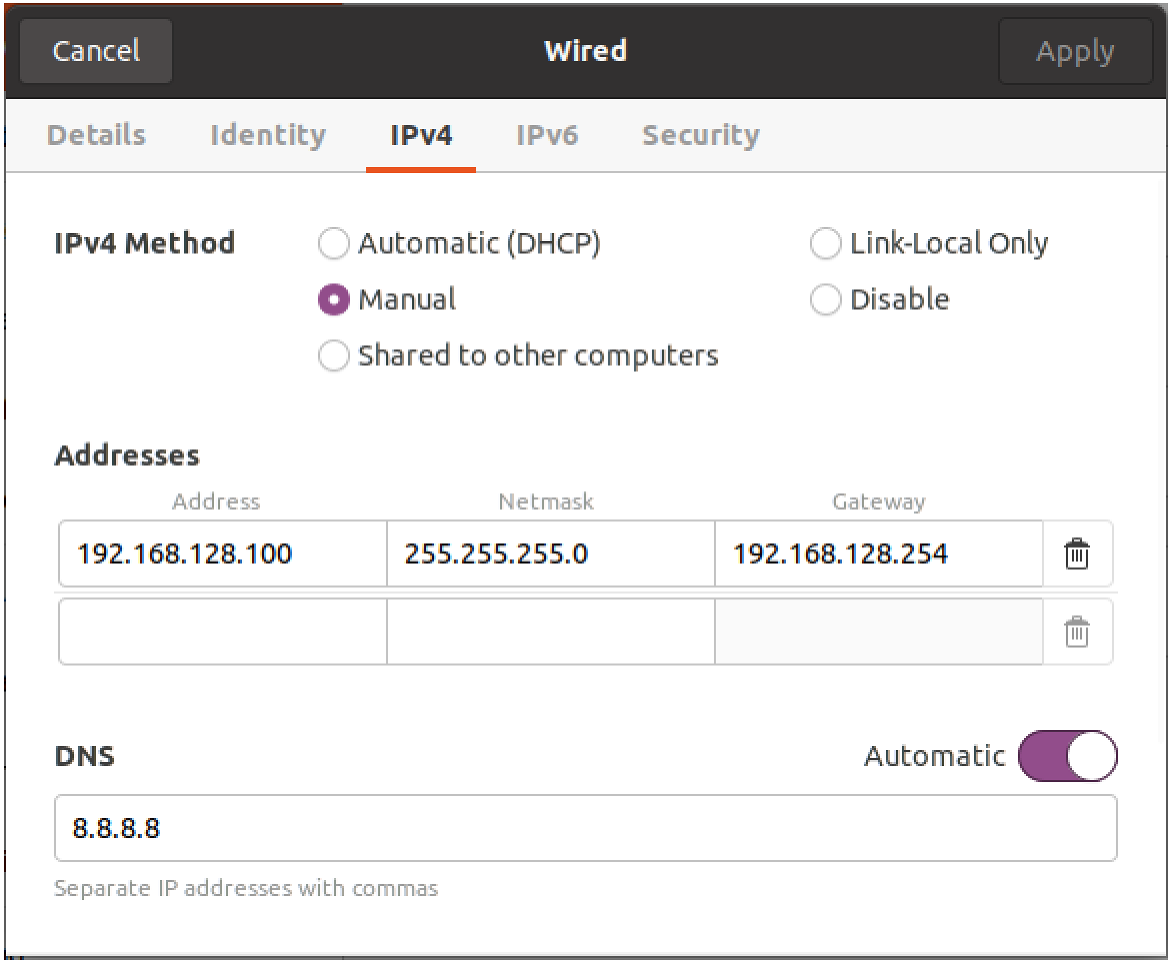
Server2 설정

- Network Adapter : Host-Only



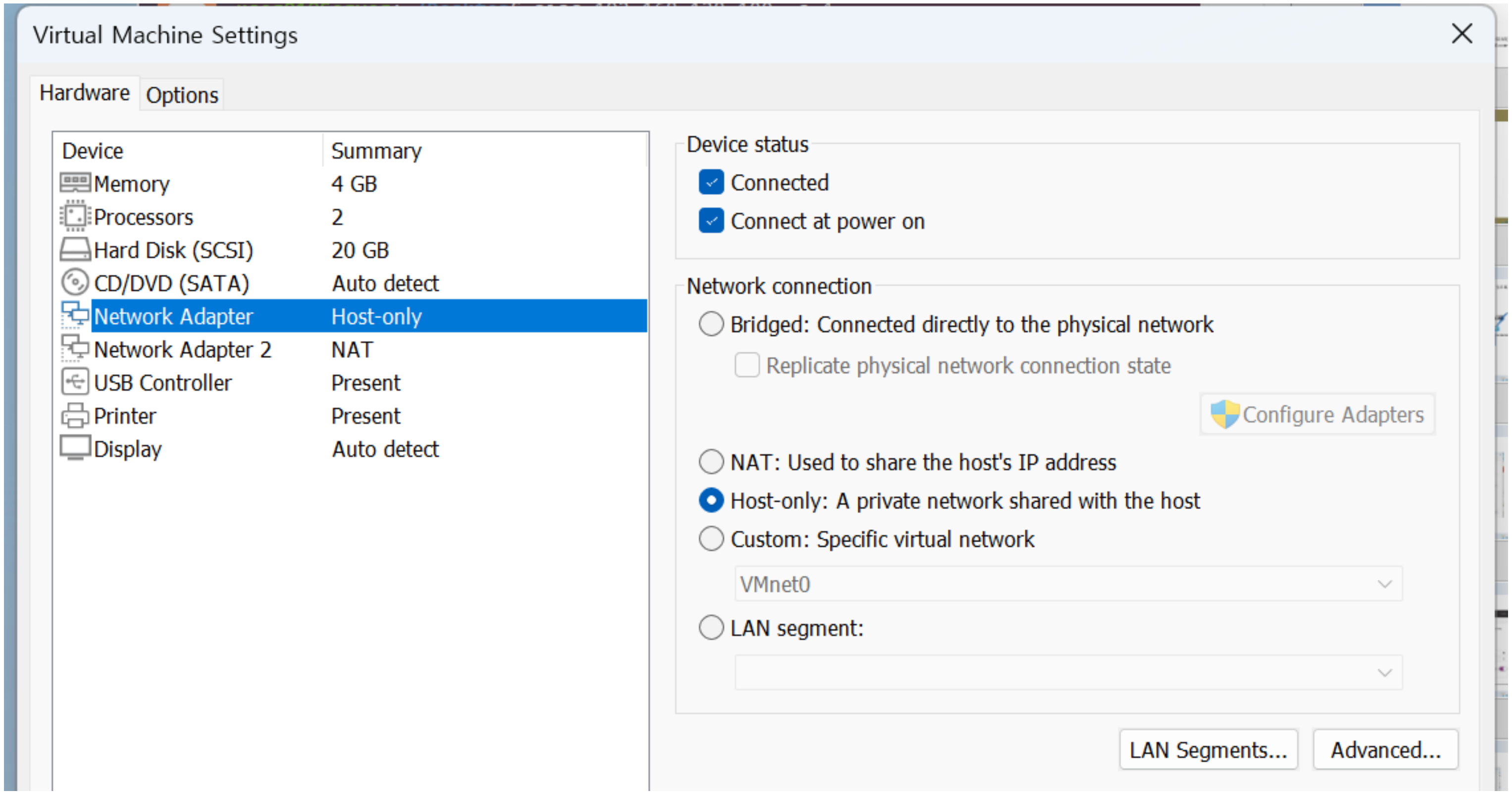
Server2 설정

- Ubuntu Network 설정
  - IPv4 Method : Manual
  - IP : 192.168.128.100
  - Netmask : 24
  - Gateway : 192.168.128.1
  - DNS : 8.8.8.8



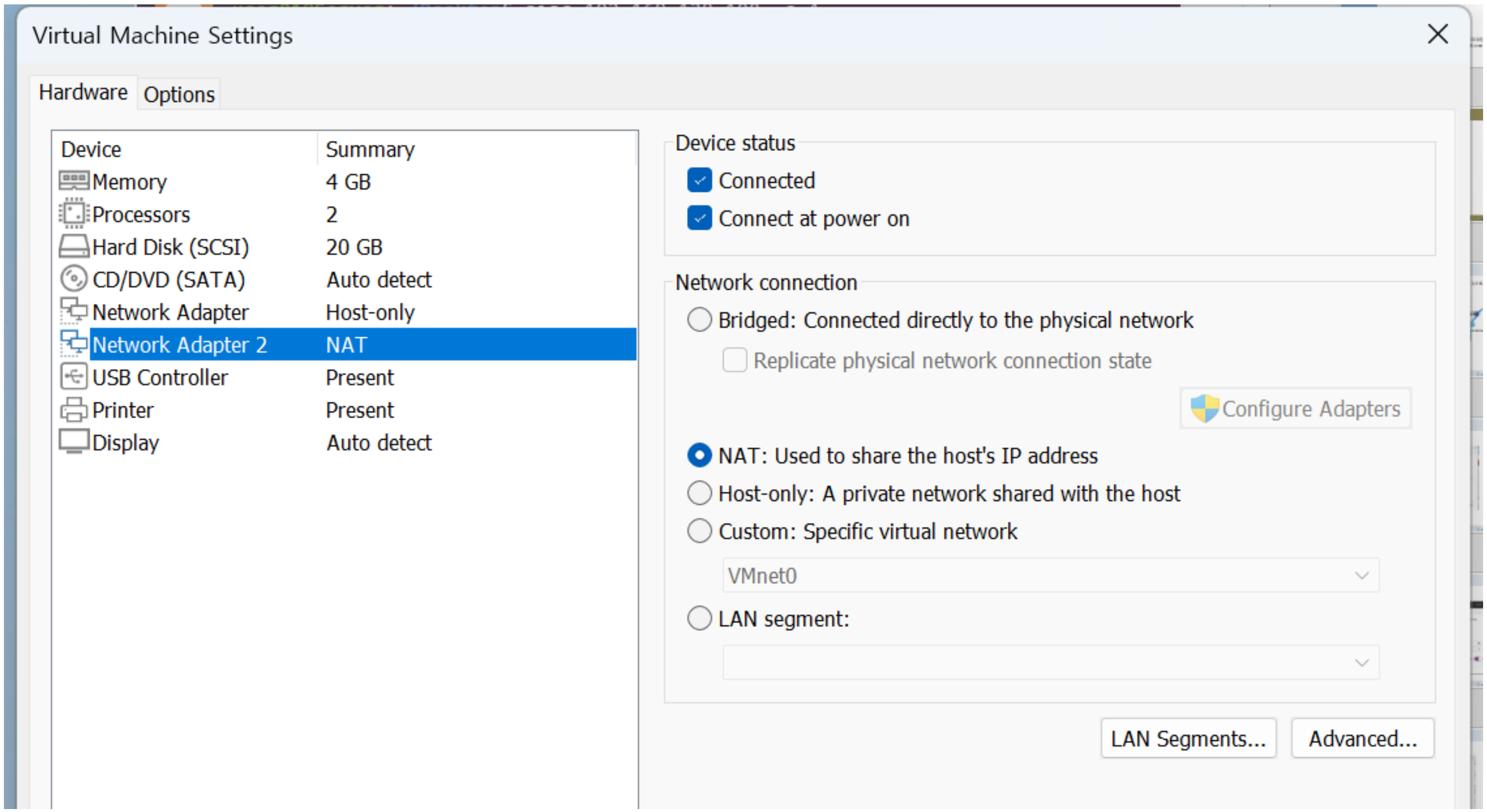
Router 설정

- Network Adapter : Host-Only



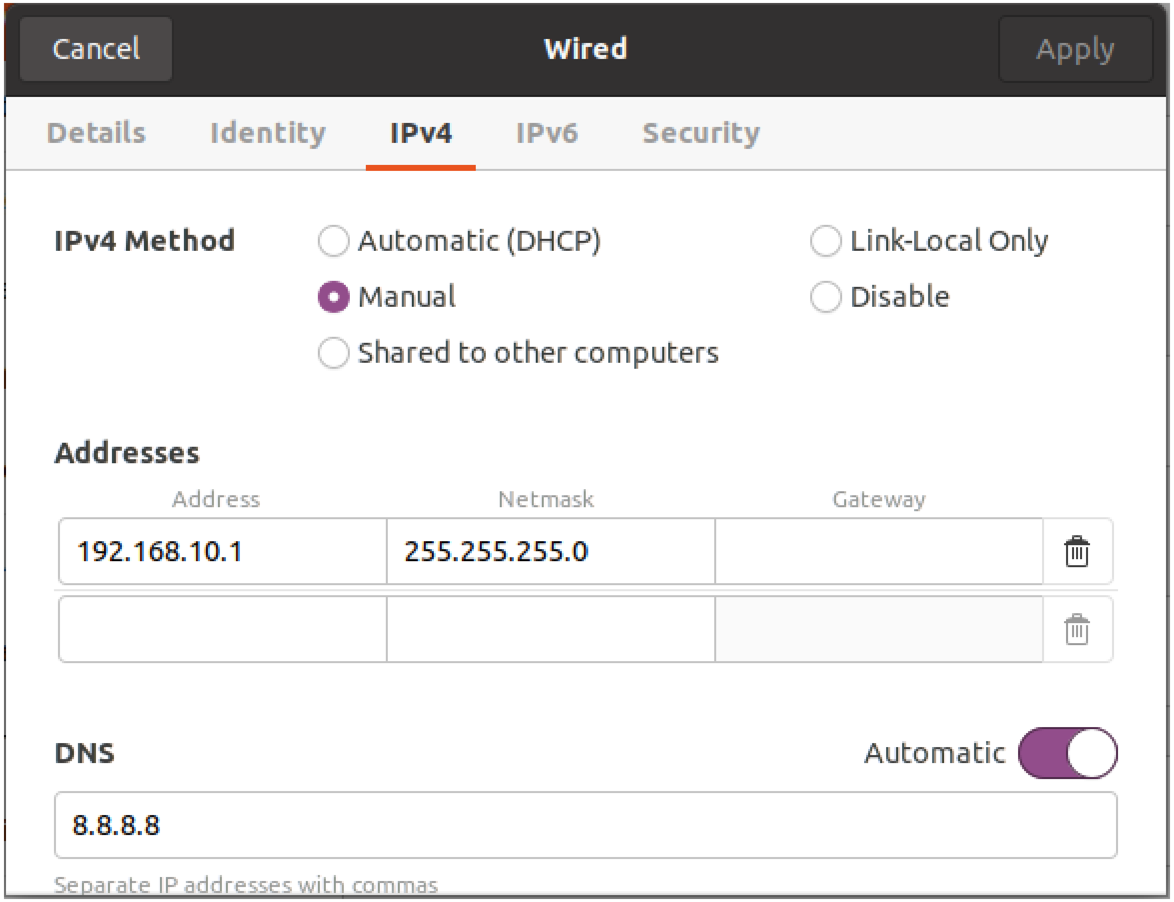
Router 설정

- Network Adapter 2 : NAT



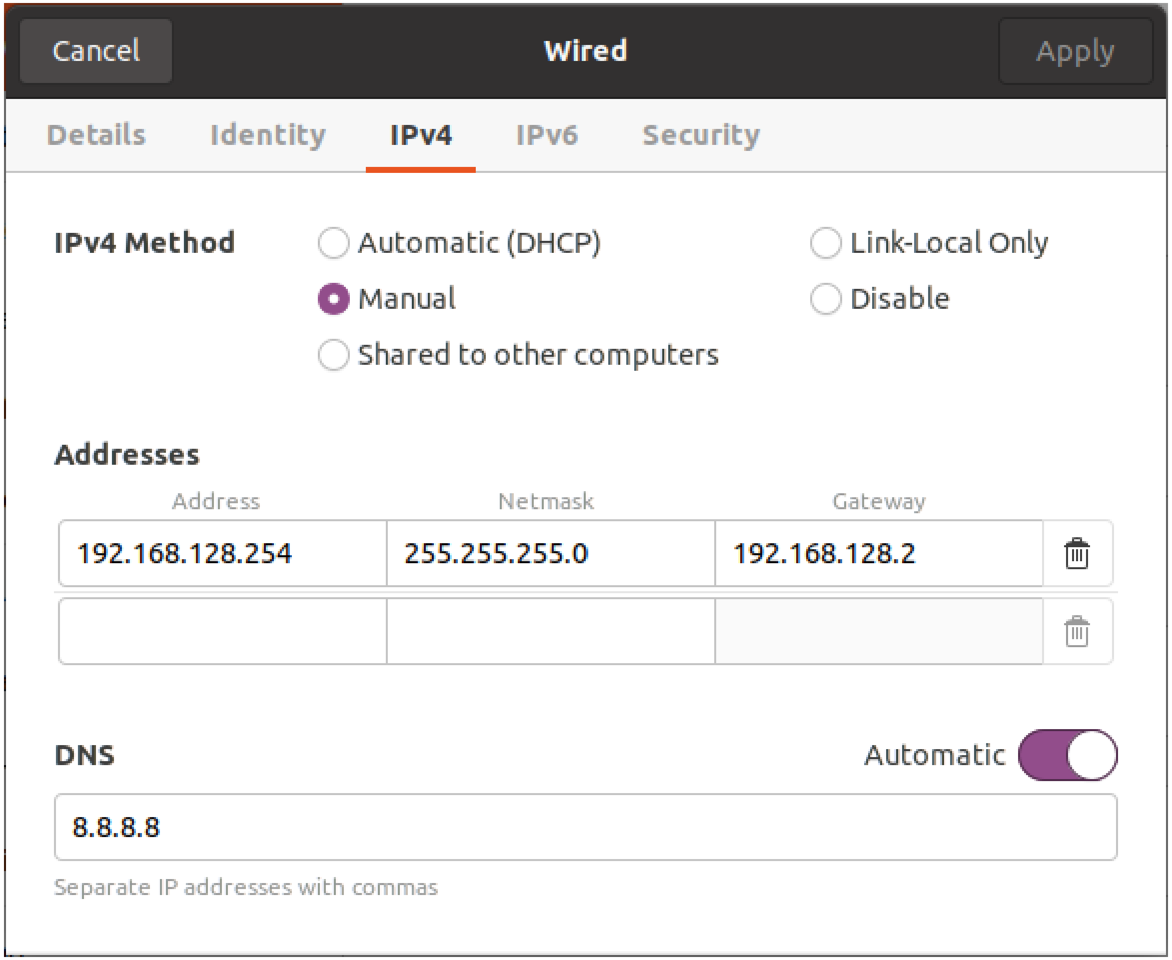
Router 설정

- Ubuntu Network 설정
  - Interface : ens33
  - IPv4 Method : Manual
  - IP : 192.168.10.1
  - Netmask : 24
  - DNS : 8.8.8.8



Router 설정

- Ubuntu Network 2 설정
  - Interface : ens35
  - IPv4 Method : Manual
  - IP : 192.168.128.254
  - Netmask : 24
  - Gateway : 192.168.128.2
  - DNS : 8.8.8.8



Router 설정

- Interface간 패킷 전달
  - ens33 Interface와 ens35 Interface간 패킷 전송을 위한 Routing 기능 활성화
  - $ sudo vi /etc/sysctl.conf

```
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=0

# Uncomment the next line to enable packet forwarding for IPv6
#  Enabling this option disables Stateless Address Autoconfiguration
#  based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

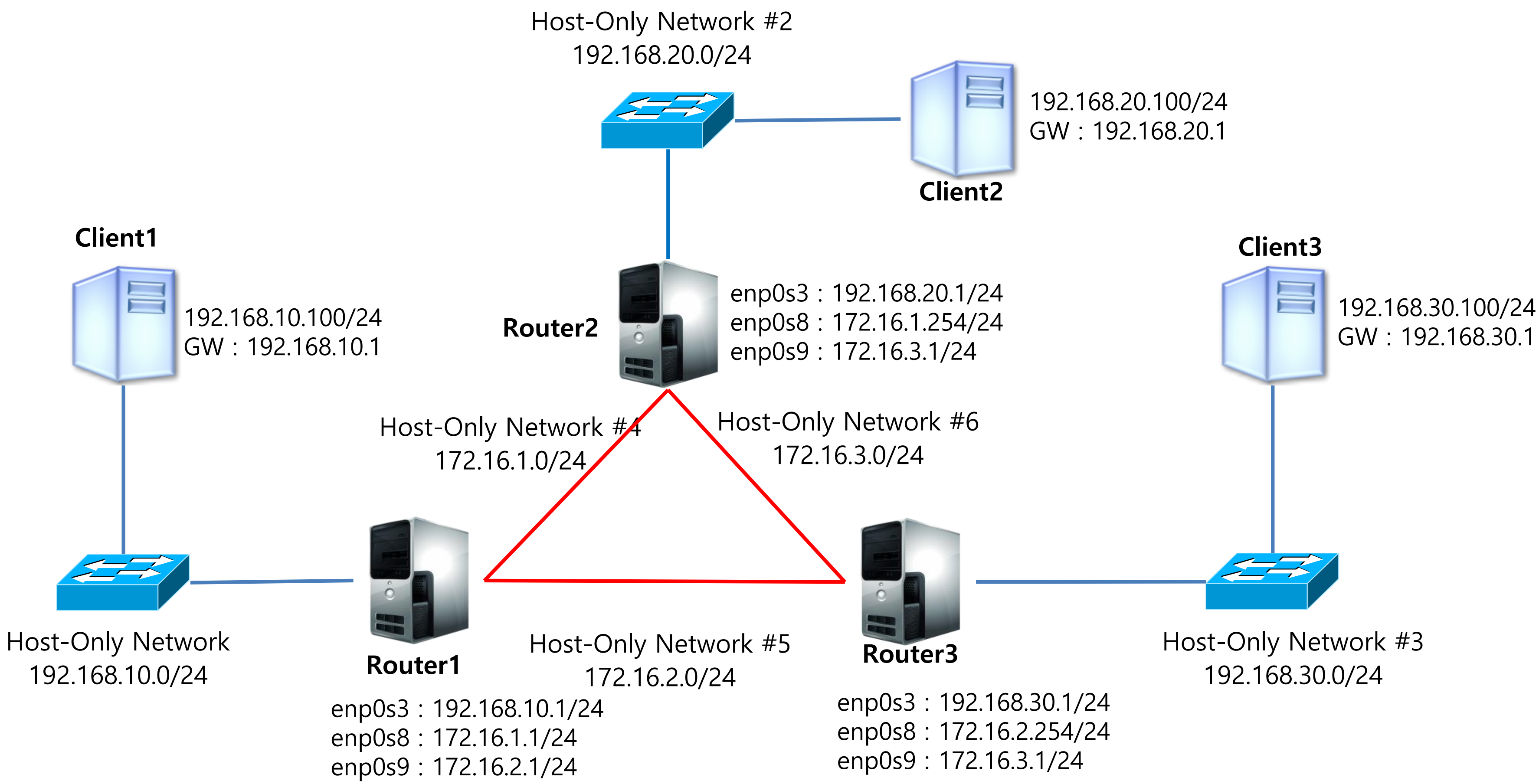
  - #net.ipv4.ip_forwarding=1 행을 찾아서 맨 앞의 # 제거 후 저장
  - 변경된 내용을 적용
  - $ sudo sysctl -p /etc/sysctl.conf

```
user01@Server:~/Desktop$ sudo sysctl -p /etc/sysctl.conf
[sudo] password for user01:
net.ipv4.ip_forward = 1
user01@Server:~/Desktop$
```

2. RIP 라우팅 연결

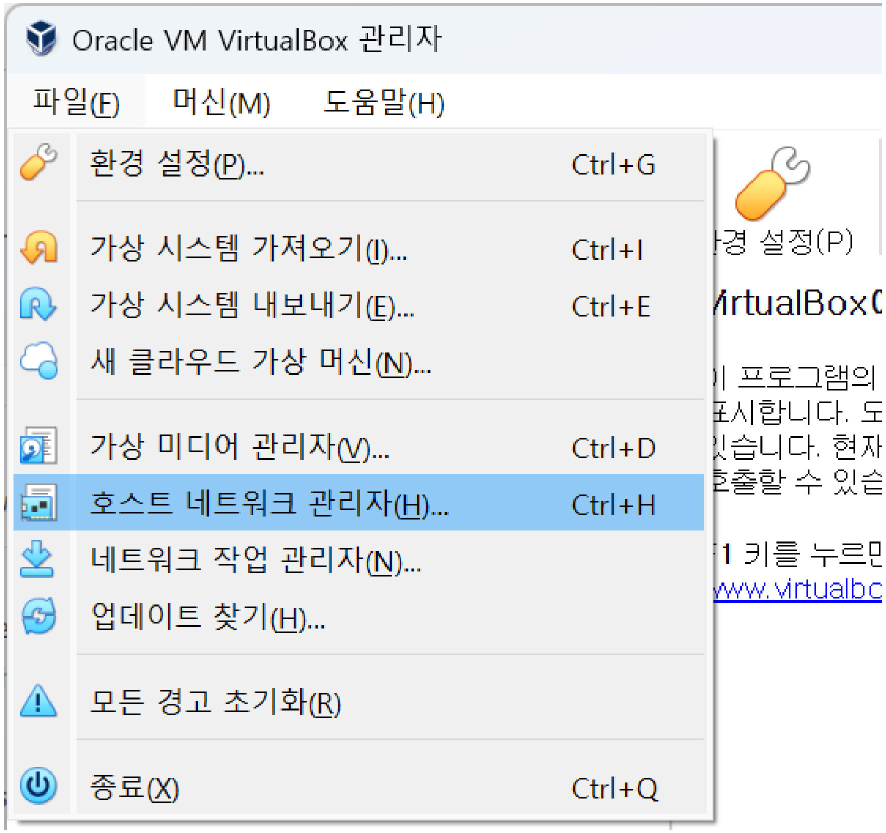
구성도

- 우분투 리눅스로 RIP라우팅 프로토콜을 실행하여 Dynamic Routing을 통해 상호 네트워크들간 통신을 할 수 있도록 설정하도록 합니다.



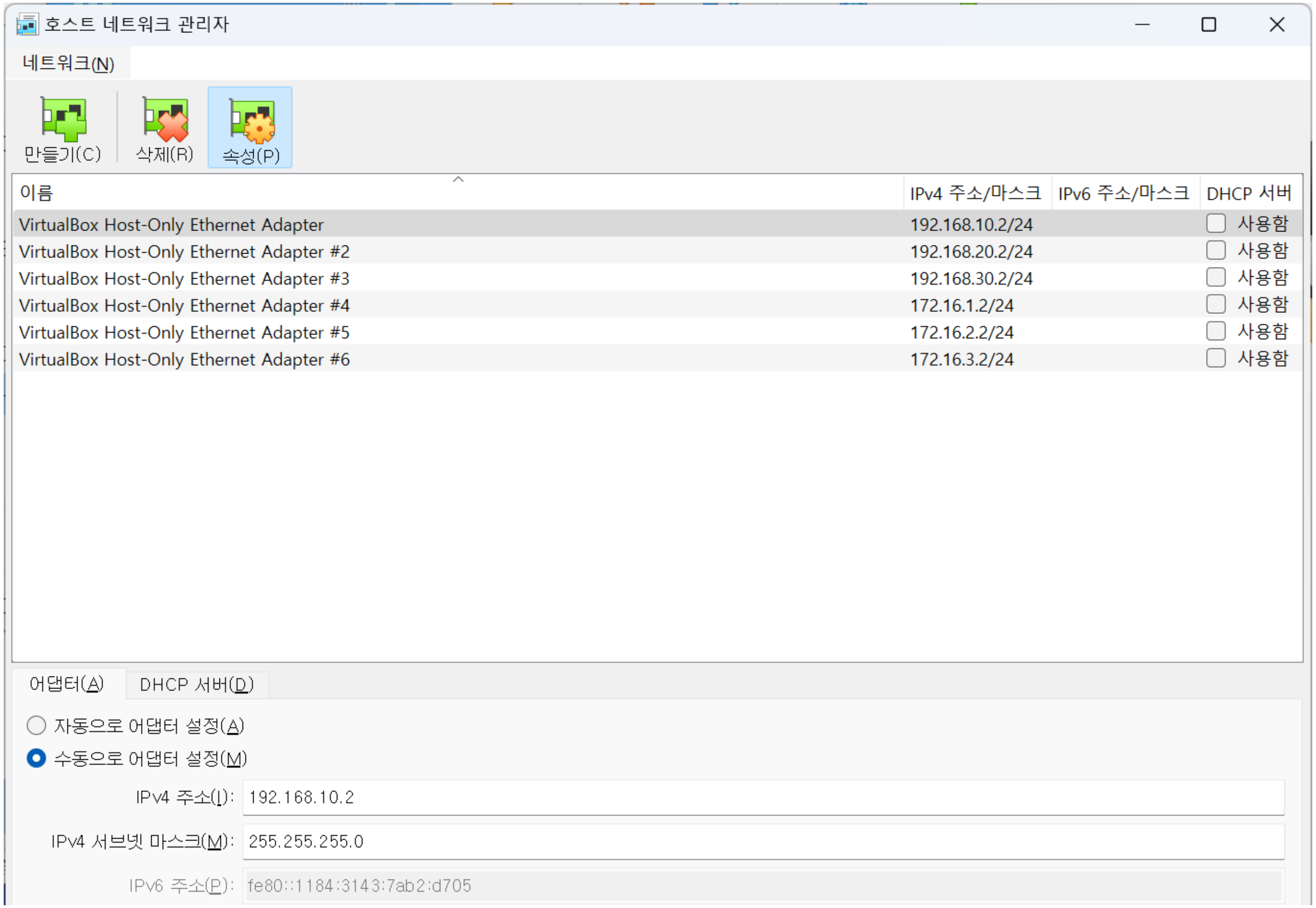
Network 환경 설정

- VirtualBox 프로그램에서 '파일(F)'→'호스트 네트워크 관리자(H)…'를 실행한다.



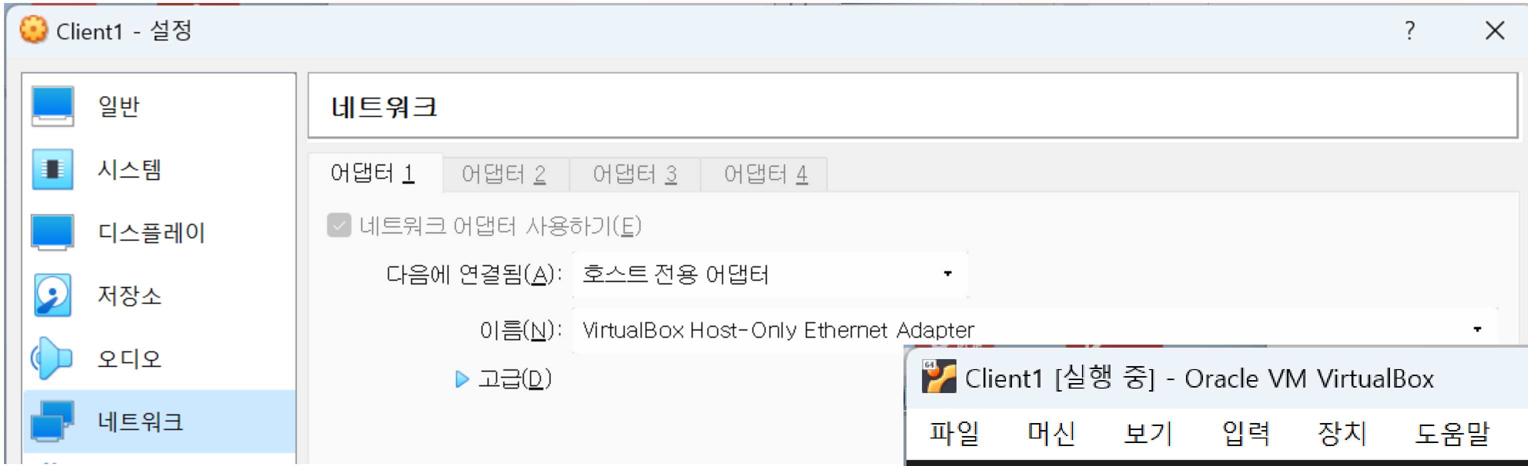
Network 환경 설정

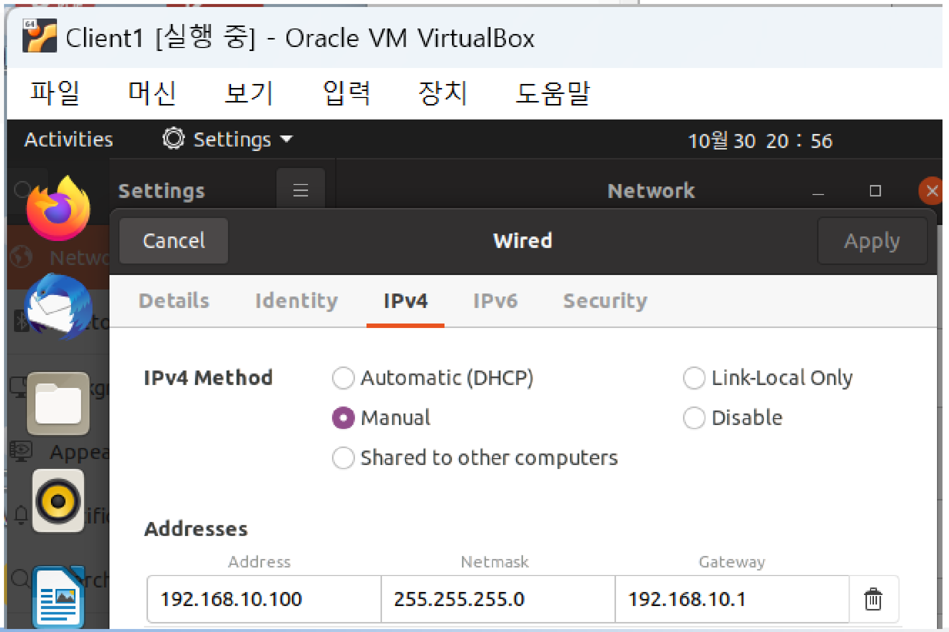
- '만들기(C)'를 클릭하여 다음과 같이 VirtualBox Host-Only Ethernet Adapter를 #6번까지 생성하고 'IPv4 주소'와 '서브넷 마스크'를 수정해 준다.



Client 설정

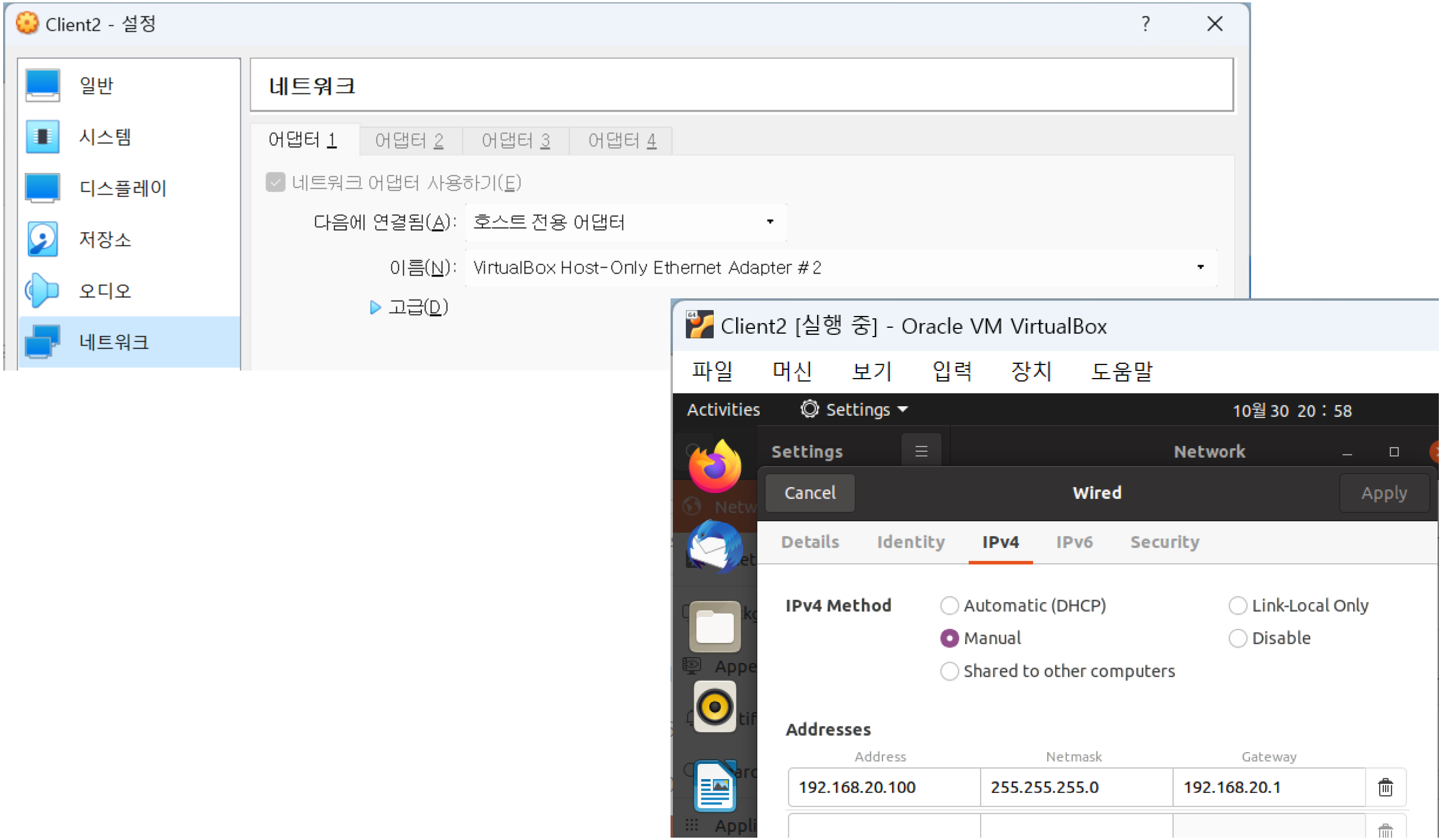
- Client1 이미지를 완전 복제로 설정하여 Client2, Client3 이미지를 생성한다.
- Client1의 정보를 다음과 같이 설정하도록 한다.





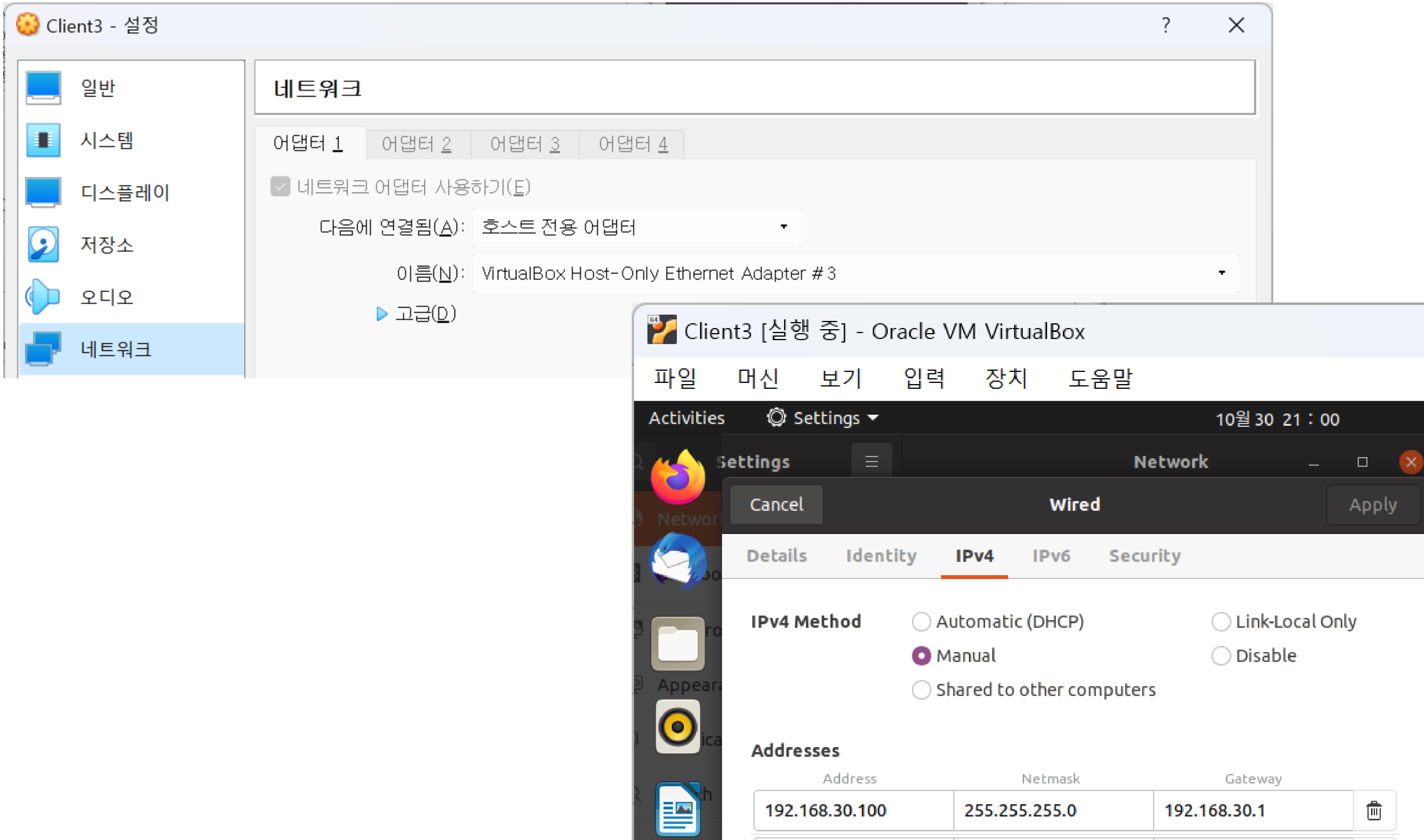
Client 설정

- Client2의 정보를 다음과 같이 설정하도록 한다.



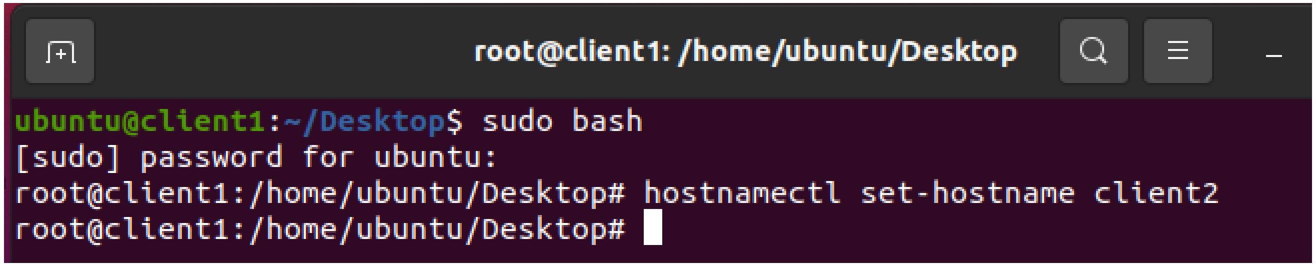
Client 설정

- Client3의 정보를 다음과 같이 설정하도록 한다.

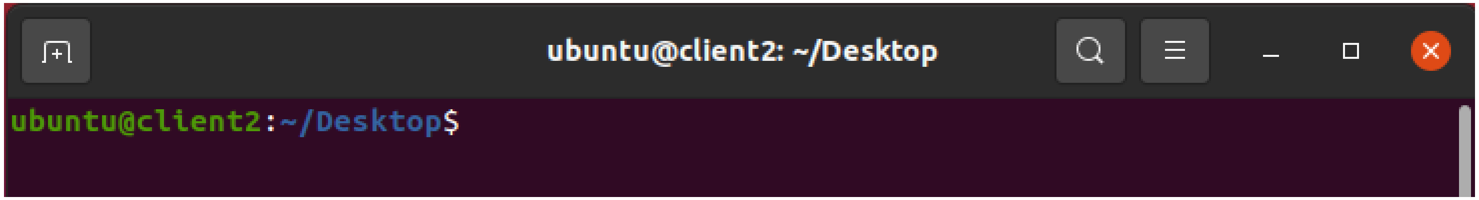


Client 설정

- Client를 실행하여 터미널창에서 Client의 이름을 변경해 준다.
  - 관리자 권한으로 변경
  - hostnamectl set-hostname 변경할이름



  - 터미널 창을 종료하고 다시 오픈하여 호스트 이름이 변경되었는지 확인한다.



Router 설정

- Router를 종료하고 다음과 같이 네트워크 카드를 2개 추가하고 호스트전용 어댑터에 배정하도록 한다.

Router 설정

- Router 이미지를 실행하고 quagga 패키지를 설치하도록 한다.
  - apt install quagga

root@router1: /home/ubuntu/Desktop

```
ubuntu@router1:~/Desktop$ sudo bash
root@router1:/home/ubuntu/Desktop# apt install quagga
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  quagga-bgpd quagga-core quagga-isisd quagga-ospf6d quagga-ospfd quagga-pimd
  quagga-ripd quagga-ripngd
Suggested packages:
  snmpd
The following NEW packages will be installed:
  quagga quagga-bgpd quagga-core quagga-isisd quagga-ospf6d quagga-ospfd
  quagga-pimd quagga-ripd quagga-ripngd
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,276 kB of archives.
After this operation, 5,716 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Router 설정

- 기본 설정파일들을 생성해 준다.
  - vi /etc/quagga/deamons

```
root@router1: /home/ubuntu/Desktop

zebra=1
ripd=1
~
```

  - vi /etc/quagga/zebra.conf

```
root@router1: /home/ubuntu/Desktop

hostname router1
password quagga
enable password quagga
~
```

- hostname 은 동일하게
- password는 임의로

  - vi /etc/quagga/ripd.conf

```
root@router1: /home/ubuntu/Desktop

hostname ripd
password quagga
~
```

Router 설정

- /etc/services 파일에 zebra와 ripd port 등록 후 재부팅
  - vi /etc/ services

```
root@router1: /home/ubuntu/Desktop

rtcm-sc104      2101/tcp                        # RTCM SC-104 IANA 1/29/99
rtcm-sc104      2101/udp
gsigatekeeper   2119/tcp
gris            2135/tcp                # Grid Resource Information Server
cvspserver      2401/tcp                        # CVS client/server operations
venus           2430/tcp                        # codacon port
venus           2430/udp                        # Venus callback/wbc interface
venus-se        2431/tcp                        # tcp side effects
venus-se        2431/udp                        # udp sftp side effect
codasrv         2432/tcp                        # not used
codasrv         2432/udp                        # server port
codasrv-se      2433/tcp                        # tcp side effects
codasrv-se      2433/udp                        # udp sftp side effect
mon             2583/tcp                        # MON traps
mon             2583/udp
zebra           2601/tcp
ripd            2602/tcp
dict            2628/tcp                        # Dictionary server
f5-globalsite   2792/tcp
gsiftp          2811/tcp
gpsd            2947/tcp
gds-db          3050/tcp        gds_db          # InterBase server
icpv2           3130/tcp        icp             # Internet Cache Protocol
icpv2           3130/udp        icp
isns            3205/tcp                        # iSNS Server Port
isns            3205/udp                        # iSNS Server Port
iscsi-target    3260/tcp
```

Router 설정

- quagga 접속 및 네트워크 정보 설정

ubuntu@router1: ~/Desktop

```
ubuntu@router1:~/Desktop$ telnet localhost zebra
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

Hello, this is Quagga (version 1.2.4).
Copyright 1996-2005 Kunihiro Ishiguro, et al.


User Access Verification

Password:
router1> en
Password:
router1# conf t
router1(config)# int enp0s3
router1(config-if)# ip address 192.168.10.1/24
router1(config-if)# exit
router1(config)# int enp0s8
router1(config-if)# ip address 172.16.1.1/24
router1(config-if)# exit
router1(config)# int enp0s9
router1(config-if)# ip address 172.16.2.1/24
router1(config-if)# end
router1# write memory
Configuration saved to /etc/quagga/zebra.conf
router1#
```

Router 설정

■ 네트워크 정보 설정 확인

```
ubuntu@router1: ~/

router1# show running-config

Current configuration:
!
hostname router1
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.10.1/24
!
interface enp0s8
 ip address 172.16.1.1/24
!
interface enp0s9
 ip address 172.16.2.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
end
router1# quit
Connection closed by foreign host.
ubuntu@router1:~/Desktop$
```

```
ubuntu@router1: ~/Desktop

ubuntu@router1:~/Desktop$ sudo cat /etc/quagga/zebra.conf
[sudo] password for ubuntu:
!
! Zebra configuration saved from vty
!   2022/10/30 21:45:47
!
hostname router1
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.10.1/24
!
interface enp0s8
 ip address 172.16.1.1/24
!
interface enp0s9
 ip address 172.16.2.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
ubuntu@router1:~/Desktop$
```

Router 설정

- Rip 프로토콜 정보 설정

```
ubuntu@router1: ~/Desktop
ubuntu@router1:~/Desktop$ telnet localhost ripd
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

Hello, this is Quagga (version 1.2.4).
Copyright 1996-2005 Kunihiro Ishiguro, et al.


User Access Verification

Password:
ripd> en
ripd# conf t
ripd(config)# router rip
ripd(config-router)# network 192.168.10.0/24
ripd(config-router)# network 172.16.1.0/24
ripd(config-router)# network 172.16.2.0/24
ripd(config-router)# end
ripd# write memory
Configuration saved to /etc/quagga/ripd.conf
ripd#
```

Router 설정

- Rip 프로토콜 정보 확인

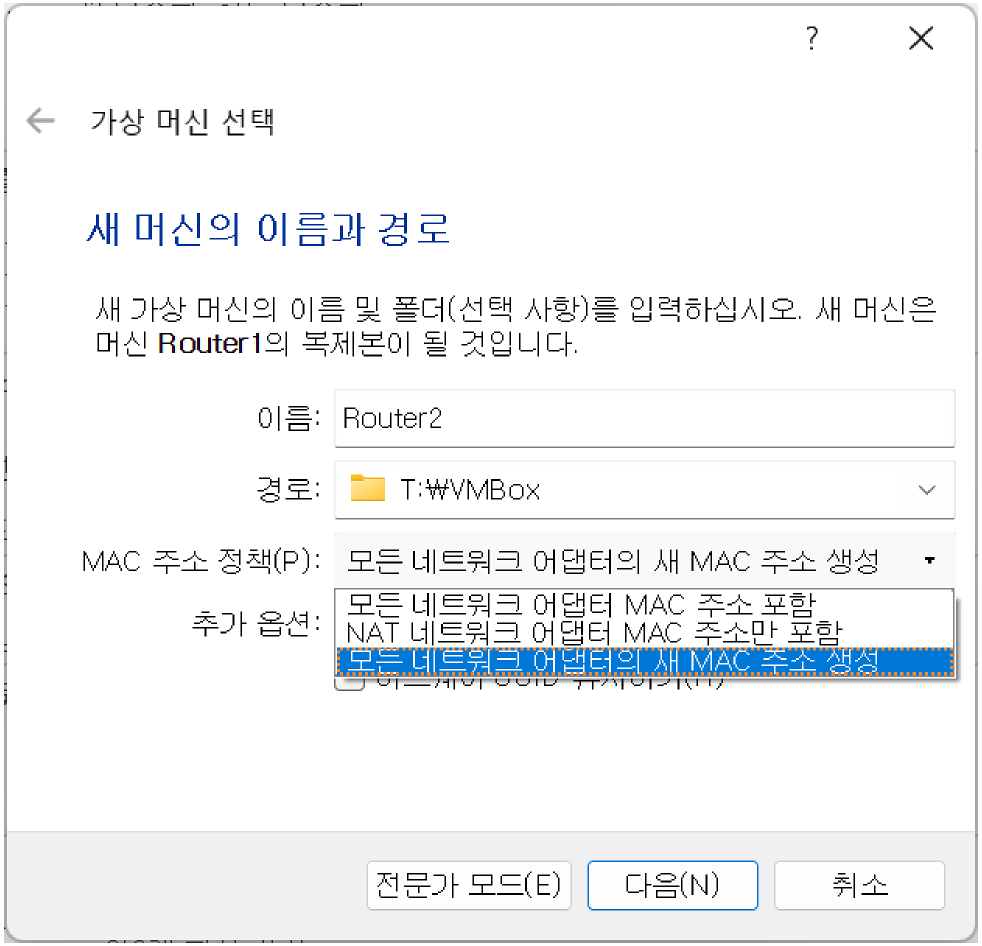
```
ripd#
ripd# show running-config

Current configuration:
!
hostname ripd
password quagga
!
router rip
 network 172.16.1.0/24
 network 172.16.2.0/24
 network 192.168.10.0/24
!
line vty
!
end
ripd# quit
Connection closed by foreign host.
ubuntu@router1:~/Desktop$
```

```
ubuntu@router1: ~/Desktop
ubuntu@router1:~/Desktop$ sudo cat /etc/quagga/ripd.conf
!
! Zebra configuration saved from vty
!   2022/10/30 21:51:41
!
hostname ripd
password quagga
!
router rip
 network 172.16.1.0/24
 network 172.16.2.0/24
 network 192.168.10.0/24
!
line vty
!
ubuntu@router1:~/Desktop$
```

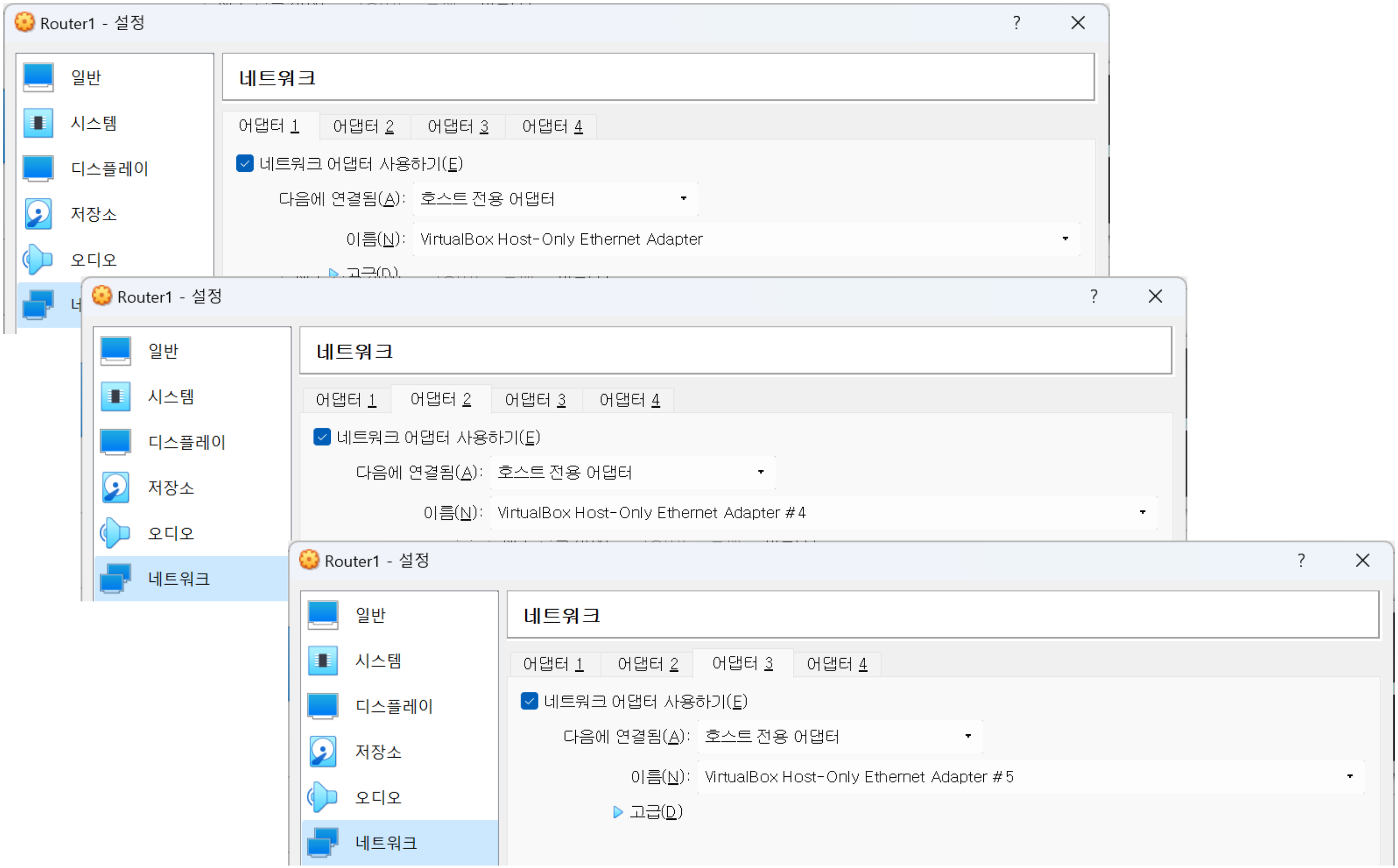
Router 설정

- OS 종료 후 Router2, Router3 이미지 복제 및 hostname 변경



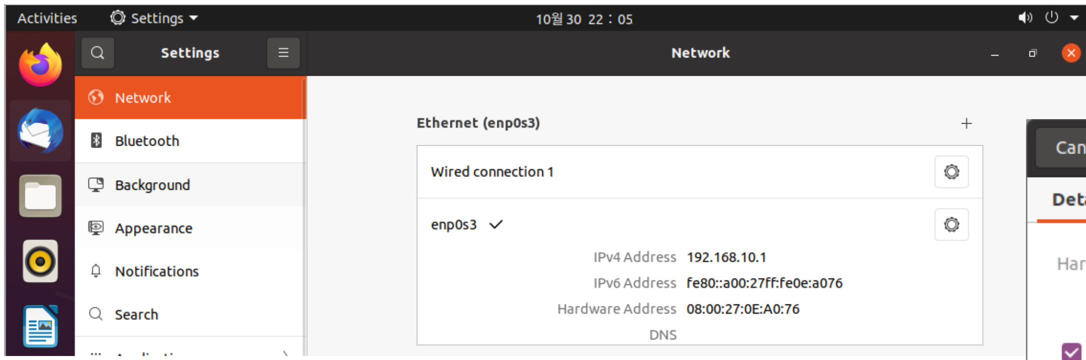
Router1 설정

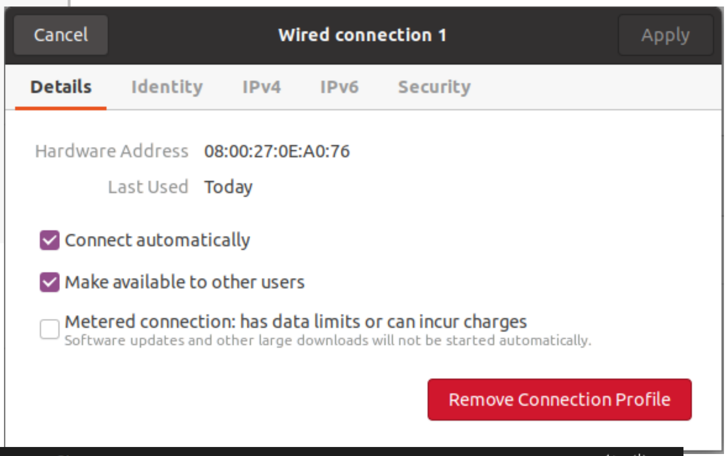
- Router1 어댑터 설정

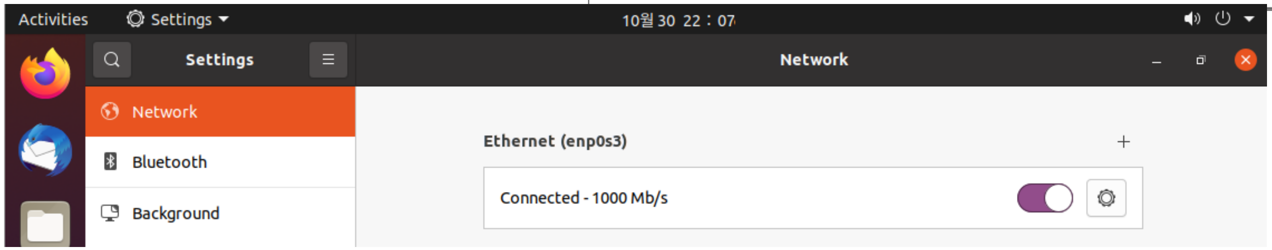


Router1 설정

- Router1 IP 설정
  - 네트워크 연결 아이콘이 나타나지 않는 경우 다음과 같이 동일한 어댑터에 여러 프로파일이 적용되어 있을 수 있음
  - '√' 기호가 없는 프로파일을 선택하여 'Remove' 시키면 된다.

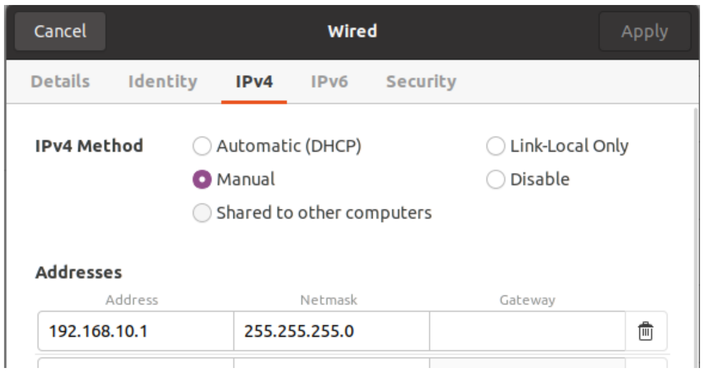




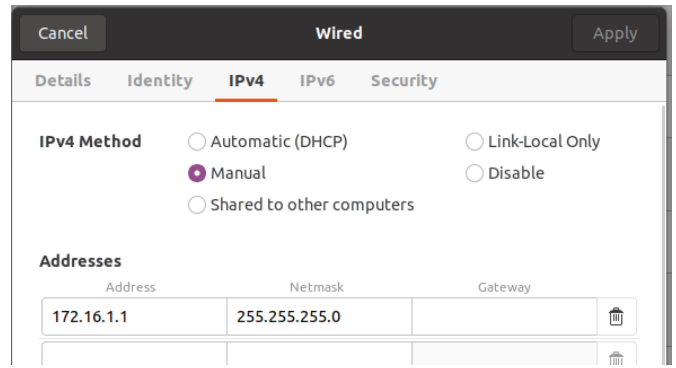


Router1 설정

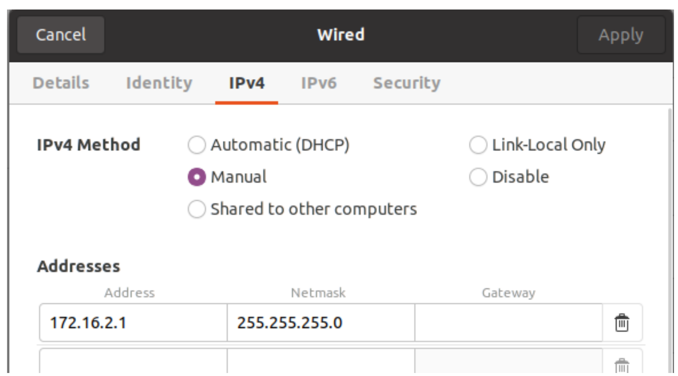
- Router1 IP 설정
  - 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.



enp0s3



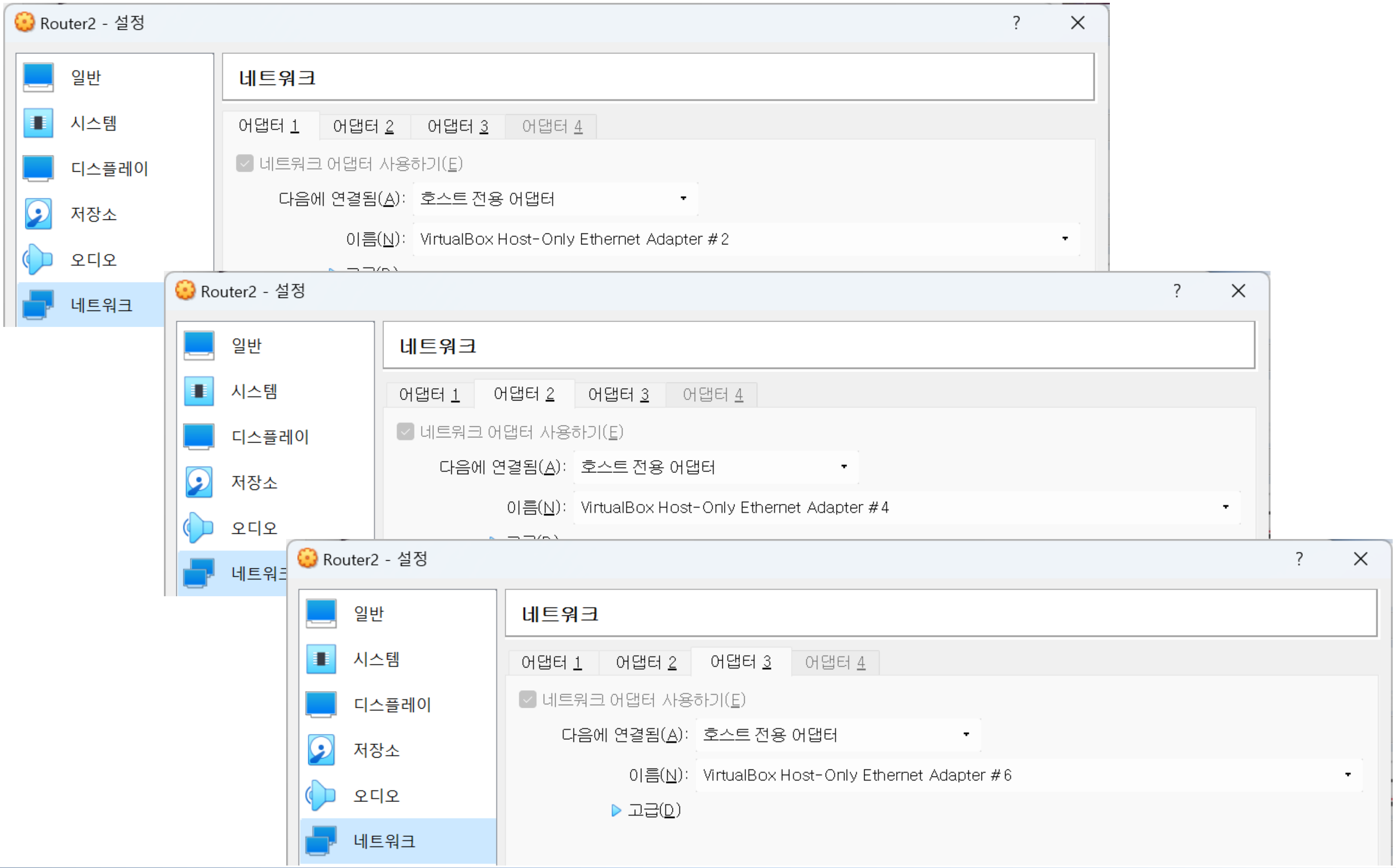
enp0s8



enp0s9

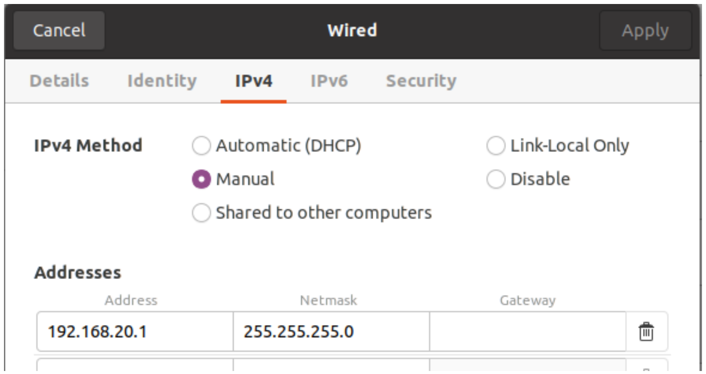
Router2 설정

Router2 어댑터 설정

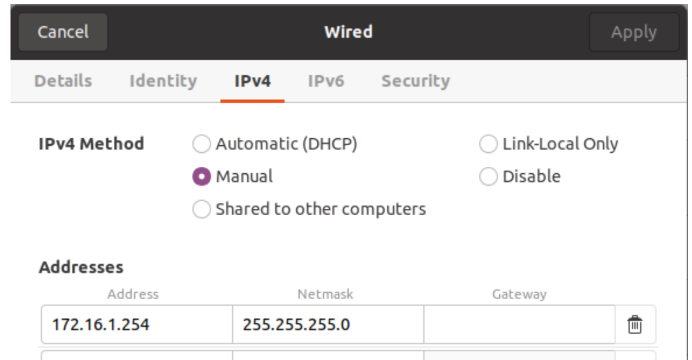


Router2 설정

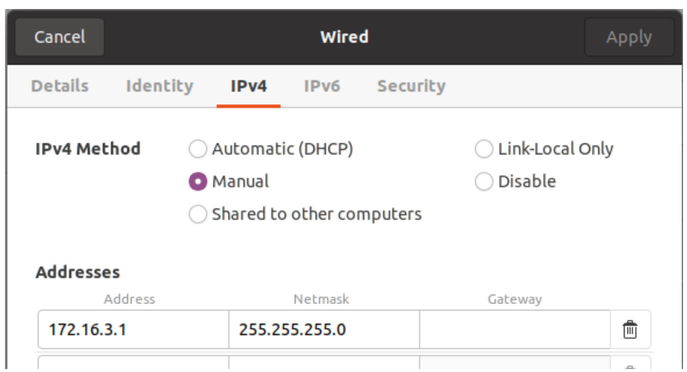
- Router2 IP 설정
  - 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.



enp0s3



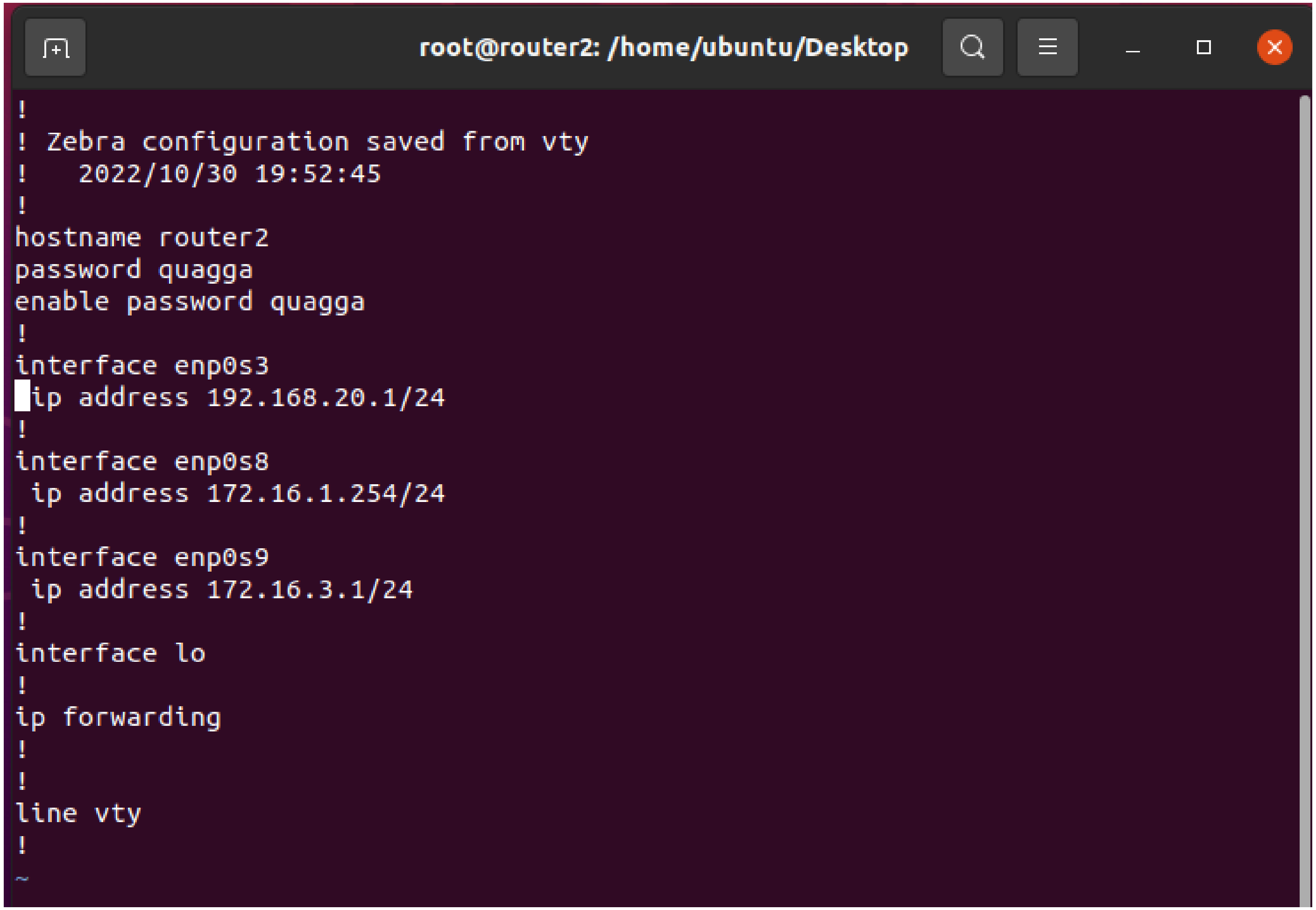
enp0s8



enp0s9

Router2 설정

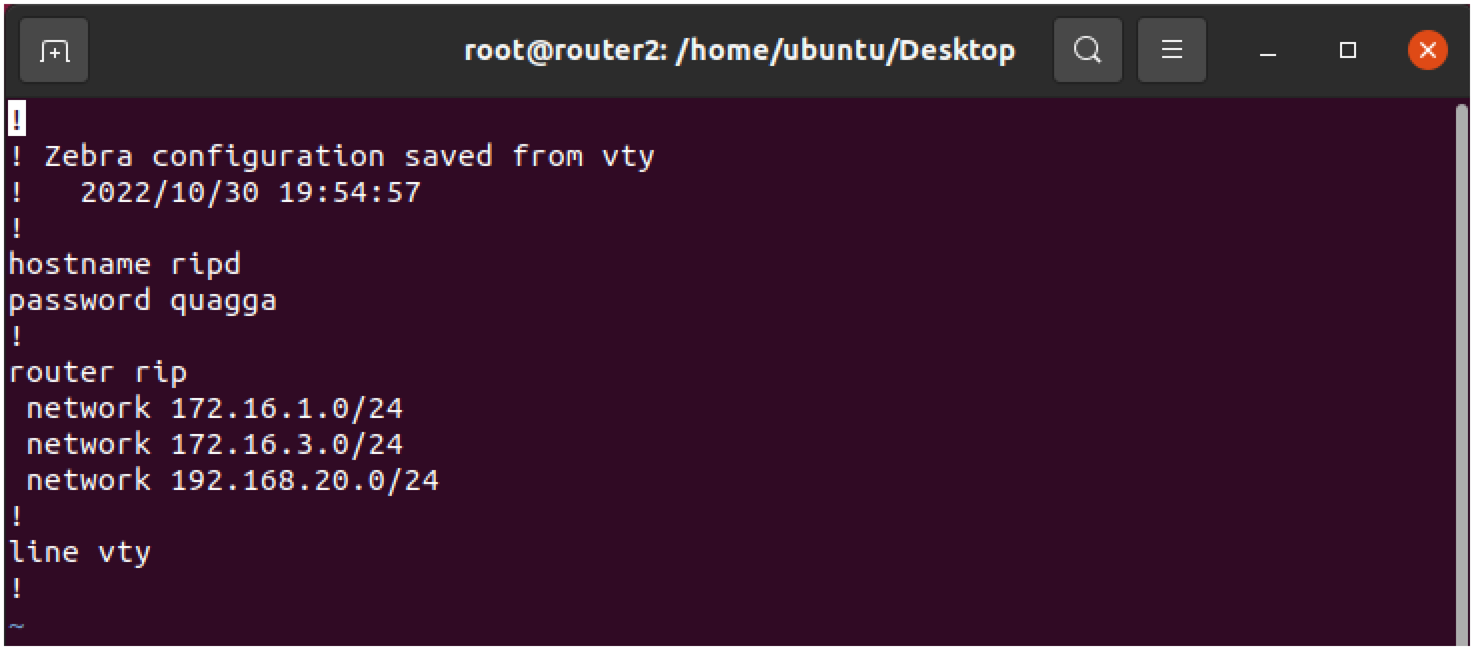
- Router2 quagga 라우터 네트워크 정보 수정
  - vi /etc/quagga/zebra.conf


root@router2: /home/ubuntu/Desktop

```
!
! Zebra configuration saved from vty
!   2022/10/30 19:52:45
!
hostname router2
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.20.1/24
!
interface enp0s8
 ip address 172.16.1.254/24
!
interface enp0s9
 ip address 172.16.3.1/24
!
interface lo
!
ip forwarding
!
!
line vty
!
~
```

Router2 설정

- Router2 Rip 프로토콜 정보 수정
  - vi /etc/quagga/ripd.conf

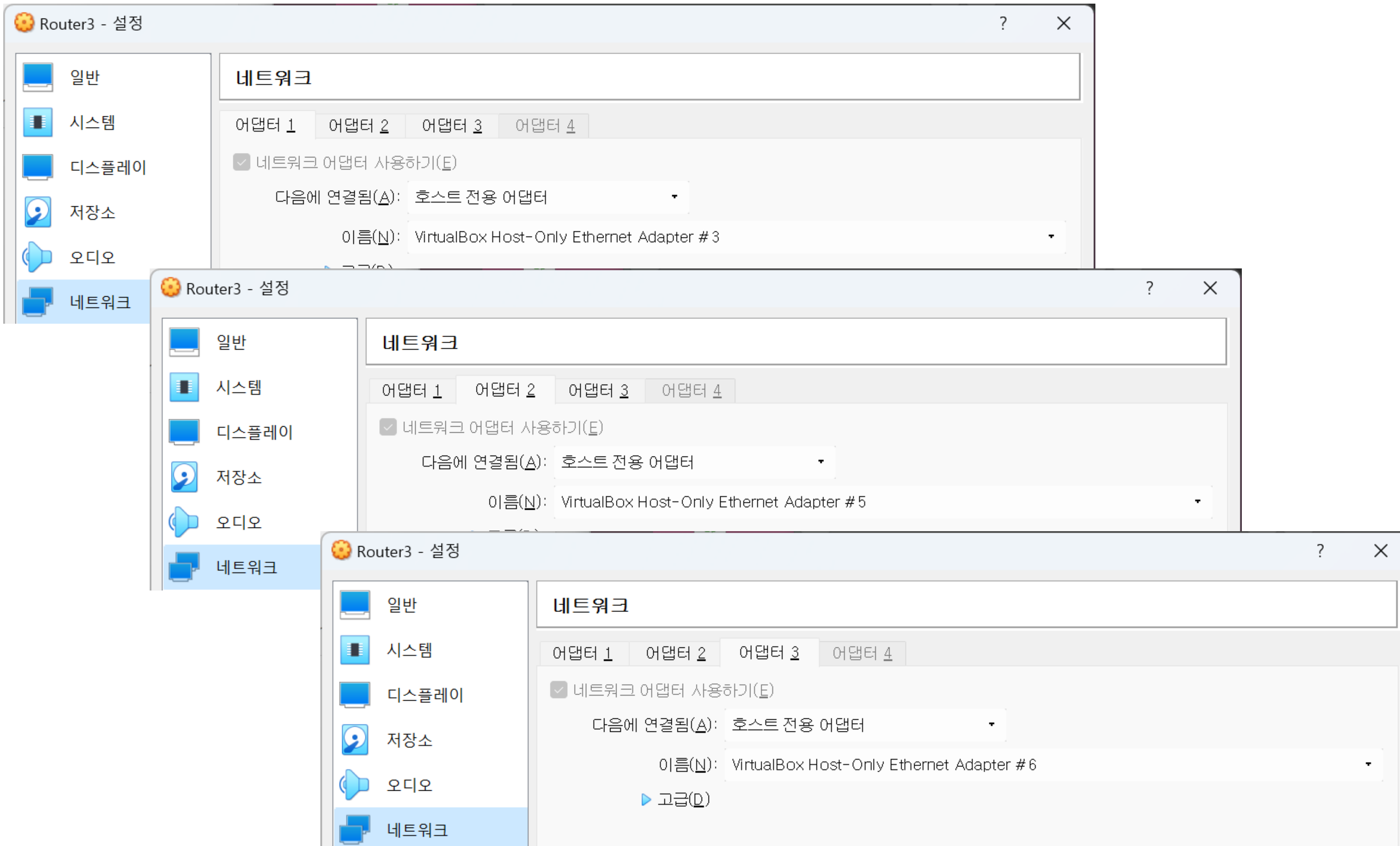


root@router2: /home/ubuntu/Desktop

```
!
! Zebra configuration saved from vty
!   2022/10/30 19:54:57
!
hostname ripd
password quagga
!
router rip
 network 172.16.1.0/24
 network 172.16.3.0/24
 network 192.168.20.0/24
!
line vty
!
~
```

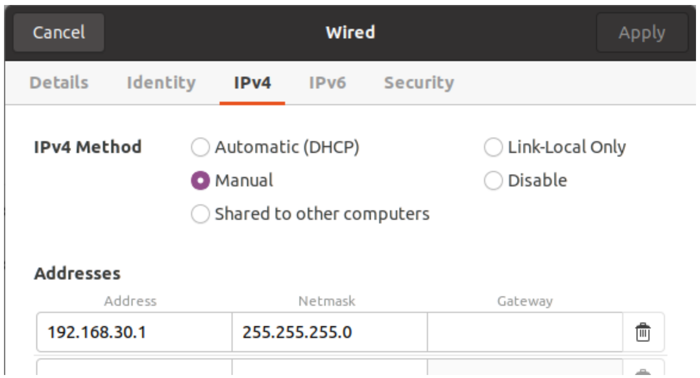
Router3 설정

- Router3 어댑터 설정

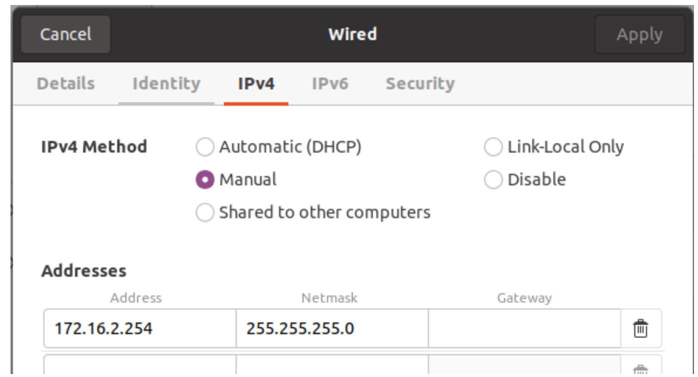


Router3 설정

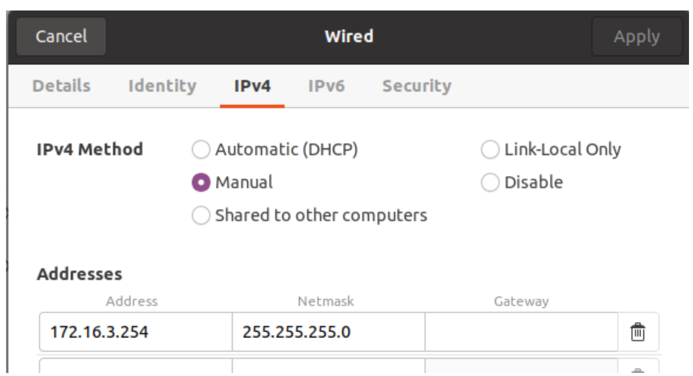
- Router3 IP 설정
  - 각각의 어댑터에 대해 다음과 같이 IP정보를 설정해 준다.



enp0s3



enp0s8



enp0s9

Router3 설정

- Router3 quagga 라우터 네트워크 정보 수정
  - vi /etc/quagga/zebra.conf

root@router3: /home/ubuntu/Desktop

```
!
! Zebra configuration saved from vty
!   2022/10/30 19:52:45
!
hostname router3
password quagga
enable password quagga
!
interface enp0s3
 ip address 192.168.30.1/24
!
interface enp0s8
 ip address 172.16.2.254/24
!
interface enp0s9
 ip address 172.16.3.254/24
!
interface lo
!
ip forwarding
!
!
line vty
!
```

Router3 설정

- Router3 Rip 프로토콜 정보 수정
  - vi /etc/quagga/ripd.conf

root@router3: /home/ubuntu/Desktop

```
!
! Zebra configuration saved from vty
!   2022/10/30 19:54:57
!
hostname ripd
password quagga
!
router rip
 network 172.16.3.0/24
 network 172.16.2.0/24
 network 192.168.30.0/24
!
line vty
!
~
```

통신 테스트

- 모든 라우터를 재부팅 하도록 한다.
- Router3 에서 라우팅 정보를 확인
  - ip route show

```
root@router3: /home/ubuntu/Desktop
root@router3:/home/ubuntu/Desktop# ip route show
169.254.0.0/16 dev enp0s3 scope link metric 1000
172.16.1.0/24 via 172.16.3.1 dev enp0s9 proto zebra metric 20
172.16.2.0/24 dev enp0s8 proto kernel scope link src 172.16.2.254
172.16.3.0/24 dev enp0s9 proto kernel scope link src 172.16.3.254
192.168.10.0/24 via 172.16.2.1 dev enp0s8 proto zebra metric 20
192.168.20.0/24 via 172.16.3.1 dev enp0s9 proto zebra metric 20
192.168.30.0/24 dev enp0s3 proto kernel scope link src 192.168.30.1
root@router3:/home/ubuntu/Desktop#
```

  - netstat -rn

```
root@router3: /home/ubuntu/Desktop
root@router3:/home/ubuntu/Desktop# netstat -rn
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
169.254.0.0     0.0.0.0         255.255.0.0     U         0 0          0 enp0s3
172.16.1.0      172.16.3.1      255.255.255.0   UG        0 0          0 enp0s9
172.16.2.0      0.0.0.0         255.255.255.0   U         0 0          0 enp0s8
172.16.3.0      0.0.0.0         255.255.255.0   U         0 0          0 enp0s9
192.168.10.0    172.16.2.1      255.255.255.0   UG        0 0          0 enp0s8
192.168.20.0    172.16.3.1      255.255.255.0   UG        0 0          0 enp0s9
192.168.30.0    0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
root@router3:/home/ubuntu/Desktop#
```

통신 테스트

- Client2 에서 ping test 수행

```
ubuntu@client2: ~/Desktop
ubuntu@client2:~/Desktop$ ping 192.168.10.100 -c 2
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=1 ttl=62 time=0.946 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=62 time=0.960 ms

--- 192.168.10.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.946/0.953/0.960/0.007 ms
ubuntu@client2:~/Desktop$ ping 192.168.30.100 -c 2
PING 192.168.30.100 (192.168.30.100) 56(84) bytes of data.
64 bytes from 192.168.30.100: icmp_seq=1 ttl=62 time=0.721 ms
64 bytes from 192.168.30.100: icmp_seq=2 ttl=62 time=0.561 ms

--- 192.168.30.100 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.561/0.641/0.721/0.080 ms
ubuntu@client2:~/Desktop$ ping 172.16.2.254 -c 2
PING 172.16.2.254 (172.16.2.254) 56(84) bytes of data.
64 bytes from 172.16.2.254: icmp_seq=1 ttl=63 time=0.835 ms
64 bytes from 172.16.2.254: icmp_seq=2 ttl=63 time=1.15 ms

--- 172.16.2.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1013ms
rtt min/avg/max/mdev = 0.835/0.990/1.146/0.155 ms
ubuntu@client2:~/Desktop$
```

End of Document